$$\mathcal{C}(\tau) = \sum_i g_i \left\langle \hat{\sigma}_z^{(i)}(\tau) \right\rangle_{\hat{\rho}\mathcal{E}} \sum_{j \neq i} g_j \left\langle \hat{\sigma}_z^{(j)} \right\rangle_{\hat{\rho}\mathcal{E}} + \sum_i g_i^2 \left\langle \hat{\sigma}_z^{(i)}(\tau) \hat{\sigma}_z^{(i)} \right\rangle_{\hat{\rho}\mathcal{E}} . \quad (5.174)$$